

Aeronautical Systems B Lecture: 4

航空宇宙システム特論B 第4回

1. Simulation of Differential Equations Using MATLAB
2. Stability Analysis

1. Matlabを使った簡単な微分方程式のシミュレーション
2. 安定解析

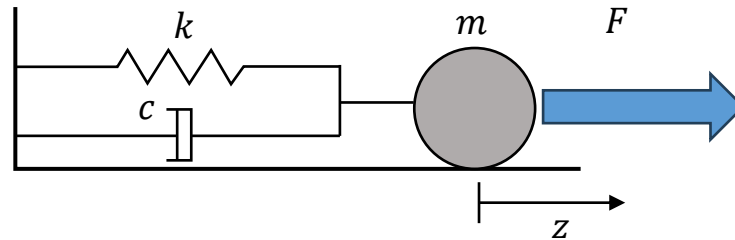


Fig 4.1 Spring-mass-damper system

$$m\ddot{z} + c\dot{z} + kz = F \quad (\text{B4-1})$$

Determine the time variation of displacement given initial values and the history of external forces.

(初期値, 外力の履歴が与えられたときの変位の時間変化を求める)

1. Expand the variables and transform it into a first-order differential equation.

(変数を拡張し 1 階の微分方程式に直す)

$$m\ddot{z} + c\dot{z} + kz = F \quad (\text{B4-1})$$

$$\ddot{z} = -\frac{c}{m}\dot{z} - \frac{k}{m}z + \frac{F}{m} \quad (\text{B4-2})$$

$$x_1 = z \quad (\text{B4-3a})$$

$$x_2 = \dot{z} \quad (\text{B4-3b})$$

Extended state variables

(拡張された状態変数)

$$\dot{x}_1 = x_2 \quad (\text{B4-4a})$$

$$\dot{x}_2 = -\frac{c}{m}x_2 - \frac{k}{m}x_1 + \frac{F}{m} \quad (\text{B4-4b})$$

First-order differential equation

(1 階の微分方程式)

2. Define the differential equation as a function and integrate it using MATLAB's differential equation solver (ode45).

(微分方程式を関数で定義し, matlabの微分方程式ソルバー(ode45)で積分する)

spring_mass_damper.m

```
function dxdt = spring_mass_damper(t,x)
```

```
x1 = x(1);
```

```
x2 = x(2);
```

```
m = 1;
```

```
c = 1;
```

```
k = 1;
```

```
F = 0;
```

```
x1dot = x2;
```

```
x2dot = -c/m*x2 - k/m*x1 - F/m;
```

```
dxdt = [x1dot  
        x2dot];
```

First derivative of the state vector

DynamicsSimulation.m

```
clear all
tspan = [0 15];
x10 = 1;
x20 = 0;

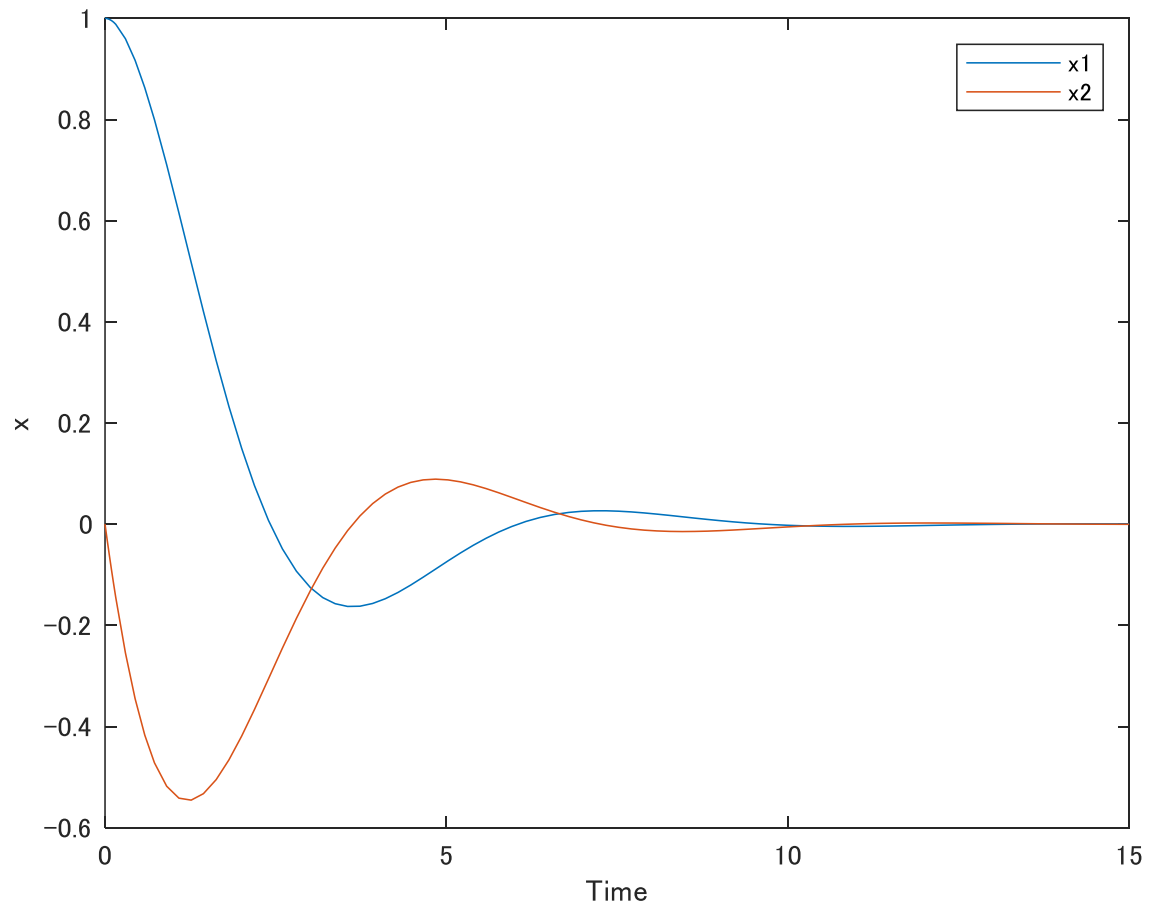
x0 = [x10
      x20];

[t,x] = ode45(@spring_mass_damper,tspan,x0);
plot(t,x)
xlabel('Time')
ylabel('x')
legend('x1','x2')
```

- ① Function name(file name) defining differential equations
- ② Integration interval
- ③ Initial state variables

Command window of matlab

```
>> DynamicsSimulation
```

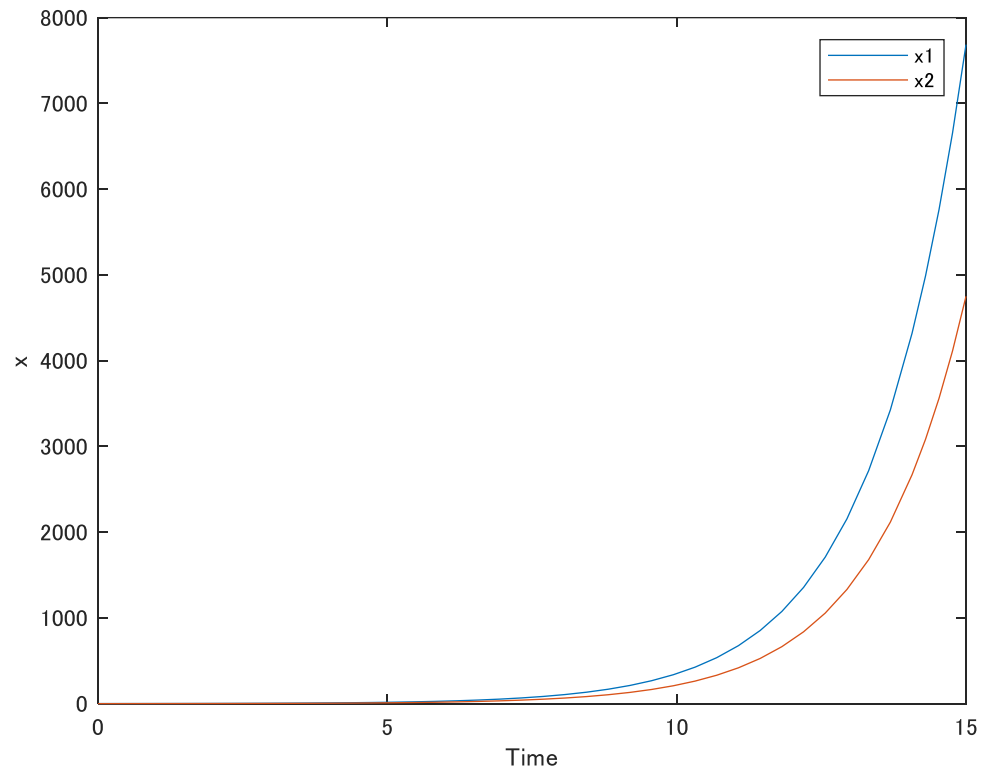


$$m\ddot{z} + c\dot{z} + kz = F$$

$$F = 0$$

$$k < 0$$

Calculate the response for
this parameter



Stable: State variables converge to an equilibrium point

Unstable: State variables diverge

安定：状態量がつり合い点に収束する

不安定：状態量が発散する

$$\dot{x}_1 = x_2 \quad (\text{B4-4a})$$

$$\dot{x}_2 = -\frac{c}{m}x_2 - \frac{k}{m}x_1 + \frac{F}{m} \quad (\text{B4-4b})$$

First-order differential equation
(1 階の微分方程式)

$$\dot{x} = Ax + Bu \quad \text{State-space representation (状態方程式)}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \text{State vector (状態ベクトル)}$$

$$u = F \quad \text{Input vector (入力ベクトル)}$$

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix} \quad \text{State(or System) matrix (システム行列)}$$

$$B = \begin{bmatrix} 0 \\ 1/m \end{bmatrix} \quad \text{Input matrix (入力行列)}$$

Stable $\Leftrightarrow$ Negative real part of system matrix eigenvalues

安定 $\Leftrightarrow$ システム行列の固有値の実部が負

AnalysisDynamics.m

```
clear all

m = 1;
c = 1;
k = -1;

A = [0 1
     -k/m -c/m];

eig(A)
```

Command window of matlab

```
>> AnalysisDynamics
```

```
ans =  
-1.6180  
0.6180
```

Unstable eigenvalue